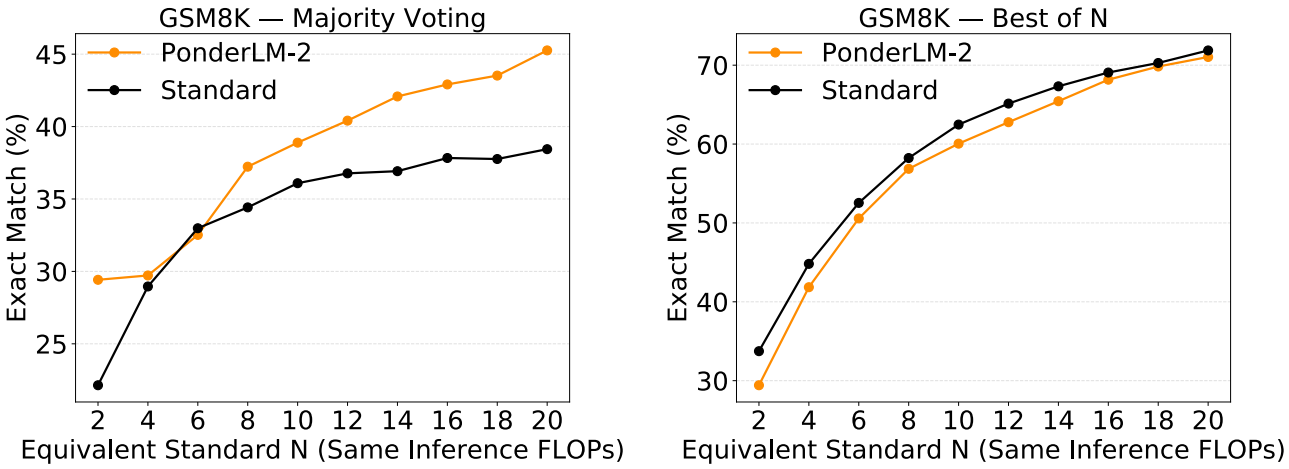
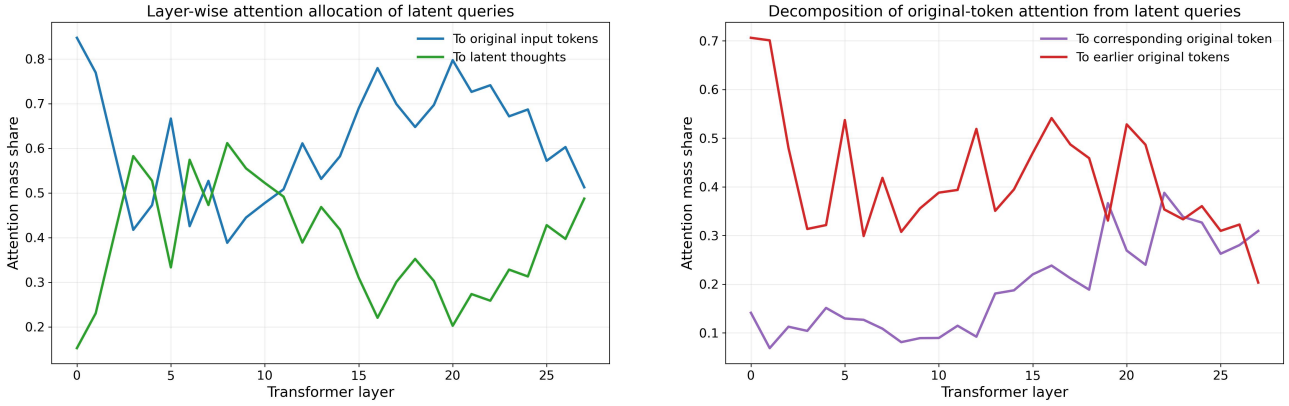


Figure 1. Compute-matched comparison of PonderLM-2 and Standard CPT on GSM8K with CoT prompting, using the LLaMA-3-3B continual-pretraining models from Section 4.4. Since one PonderLM-2 sample costs roughly **2**$\times$ a Standard CPT sample, we compare **PonderLM-2 with** N **samples** against **Standard CPT with** $2N$ **samples**, thereby matching inference FLOPs. **Left:** majority voting. **Right:** best-of-N. Under this fair compute-matched comparison, PonderLM-2 is consistently better for majority voting and remains comparable for best-of-N.



(a) Attention allocation of latent queries.

(b) Decomposition of original-token attention.

Figure 2. **Attention patterns of latent thought queries in the final forward pass** of the LLaMA-3-3B CPT model, measured across 64 sequences (2048 tokens each, excluding the first two positions to remove attention sink effects). **(a)** Layer-wise attention mass from latent queries to original input tokens vs. other latent thoughts. Lower and upper layers direct 60–80% of attention to original tokens, while middle layers split attention roughly evenly. **(b)** Decomposition of the original-token attention: lower layers broadly attend to earlier positions (context aggregation), while upper layers increasingly focus on the corresponding position's own original token (local refinement).